

Experiment - 9 Subnetting in Cisco Packet Tracer

Aim:

Implementation of Subnetting in
Cisco Packet Tracer Simulator.

CONCEPT:

Classical IP Subnetting allows
dividing a network into smaller subnets
for efficient IP address utilization. Instead
of fixed classful masks, Subnetting uses variable
-length subnet masks (VLSM) to create network
as per requirements.

STEPS:

1) Create Topology: add routers, switches and
PCs in Cisco Packet Tracer.

2) Subnetting: eg) 192.168.1.0/24 Subnetted into 10
gives 8 subnets with 30 hosts each

3) Assign IPs: Allocate Subnet IPs to routers
switches and PCs.

4) Configure Devices

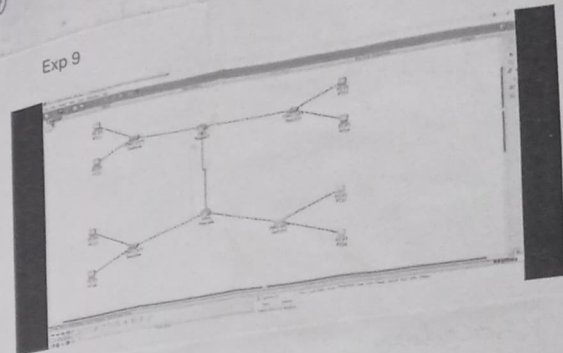
Routers → configure interface with IP &
subnet mask.

Switch → Set access mode on ports

PCs → Assign IP, Subnet mask and
default gateway.

5) Test: use Ping to check connectivity

among devices.



Student observation:

a) While doing your understanding of
Subnetting.

Ans: Subnetting divides a big network
into small part for better control.

b) What is the advantage of implementing Subnetting
within a network?

A) It improves Security, Speed and Saves IP addresses.

c) Find out whether Subnetting is implemented in
your college.

A) Yes, colleges use Subnetting

*Admin: 192.168.1.0/26

*Lab: 192.168.1.64/26

Result: Implement of Subnetting in Cisco
Packet Tracer Simulator is done Successfully.